```
syms dw1 dw2 dw3 Omega I1 I2 I3

% spin about b1
I_mat = diag([I1, I2, I3]);
omega_vec = [Omega + dw1; dw2; dw3];
H_b = I_mat * omega_vec;

pretty(H_b);
```

$$\begin{pmatrix} I_1\ (\Omega + \mathrm{dw}_1) \\ I_2\,\mathrm{dw}_2 \\ I_3\,\mathrm{dw}_3 \end{pmatrix}$$

```
dw_i = I_mat \ cross(-omega_vec, H_b);
pretty(dw_i);
```

$$\begin{pmatrix} \dfrac{I_2\,\mathrm{dw}_2\,\mathrm{dw}_3 - I_3\,\mathrm{dw}_2\,\mathrm{dw}_3}{I_1} \\ -\dfrac{I_1\,\Omega\,\mathrm{dw}_3 - I_3\,\Omega\,\mathrm{dw}_3 + I_1\,\mathrm{dw}_1\,\mathrm{dw}_3 - I_3\,\mathrm{dw}_1\,\mathrm{dw}_3}{I_2} \\ \dfrac{I_1\,\Omega\,\mathrm{dw}_2 - I_2\,\Omega\,\mathrm{dw}_2 + I_1\,\mathrm{dw}_1\,\mathrm{dw}_2 - I_2\,\mathrm{dw}_1\,\mathrm{dw}_2}{I_3} \end{pmatrix}$$

```
dw_i = subs(dw_i, [dw1*dw2, dw1*dw3, dw2*dw3], [0, 0, 0]);
pretty(dw_i);
```

$$\begin{pmatrix} 0 \\ -\dfrac{I_1\,\Omega\,\mathrm{dw}_3 - I_3\,\Omega\,\mathrm{dw}_3}{I_2} \\ \dfrac{I_1\,\Omega\,\mathrm{dw}_2 - I_2\,\Omega\,\mathrm{dw}_2}{I_3} \end{pmatrix}$$

```
dw_i = simplify(dw_i);
pretty(dw_i);
```

$$\begin{pmatrix} 0 \\ -\dfrac{\Omega\,\mathrm{dw}_3\ (I_1 - I_3)}{I_2} \\ \dfrac{\Omega\,\mathrm{dw}_2\ (I_1 - I_2)}{I_3} \end{pmatrix}$$

```
dw_i = subs(dw_i, [I1, I2, I3, Omega], [120, 100, 80, 1])
```

```
dw_i =
```

$$\begin{pmatrix} 0 \\ -\frac{2\,dw_3}{5} \\ \frac{dw_2}{4} \end{pmatrix}$$

```
pretty(dw_i)
```

$$\begin{pmatrix} 0 \\ -\frac{2\,dw_3}{5} \\ \frac{dw_2}{4} \end{pmatrix}$$

```
dw_i = simplify(dw_i);
zeroRows = all(isAlways(dw_i == 0), 2);
```

```
Warning: Unable to prove 'dw2/4 == 0'.
Warning: Unable to prove '-(2*dw3)/5 == 0'.
```

```
dw_i = dw_i(~zeroRows, :);
pretty(dw_i)
```

$$\begin{pmatrix} -\frac{2\,dw_3}{5} \\ \frac{dw_2}{4} \end{pmatrix}$$

```
x1 = [dw2; dw3]
```

x1 =

$$\begin{pmatrix} dw_2 \\ dw_3 \end{pmatrix}$$

```
A = jacobian(dw_i, x1)
```

A =

$$\begin{pmatrix} 0 & -\frac{2}{5} \\ \frac{1}{4} & 0 \end{pmatrix}$$

```
[V, D] = eigs(double(A))
```

```
V = 2×2 complex
    0.7845          +     0.0000i      0.7845          +     0.0000i
    0.0000          -     0.6202i      0.0000          +     0.6202i
D = 2×2 complex
```

```
   0.0000          +     0.3162i      0.0000          +     0.0000i
   0.0000          +     0.0000i      0.0000          -     0.3162i
```

```
answer = [1i/sqrt(10); -1i/sqrt(10)];
matlab = diag(D);

err = norm(matlab - answer)
```

```
err =
0
```

```
assert(err < 1e-12, 'incorrect eigenvalues')
disp("eigenvalues are the same");
```

```
eigenvalues are the same
```

```
V = V ./ V(1,1)
```

```
V = 2×2 complex
   1.0000          +     0.0000i      1.0000          +     0.0000i
   0.0000          -     0.7906i      0.0000          +     0.7906i
```

```
answer = [1, 1; -10j/(4*sqrt(10)), 10j/(4*sqrt(10))];
matlab = V;
err = norm(matlab - answer);
assert(err < 1e-12, 'incorrect eigenvectors')
disp("eigenvectors are the same");
```

```
eigenvectors are the same
```